1-amaliy mashg'ulot.

Matritsalarning turlari. Matritsalarni songa ko'paytirish, matritsalarni qo'shish, matritsalarni ayirish va matritsani matritsaga ko'paytirish amallari, hamda ularning xossalari.

m ta satr va n ta ustundan iborat to`g`ri burchaki ushbu

$$A=\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}=\left(a_{ij}\right)_{m\times n} \qquad (1.1)$$

jadvalga $m\times n$ o`lcovli *matritsa* deyiladi. Bu yerda $a_{ij}$ $(i=1,2,\dots,m;\ j=1,2,\dots,n)$ - matritsaning elementlari deyiladi. Matritsalar lotin alifbosidagi bosh harflar bilan belgilanadi. Ba'zan, o`lchamlarini ifodalash uchun $A_{m\times n}$ kabi belgilanadi.

Matritsalar qisqacha,

$$A=\left(a_{ij}\right)\ \ (i=\overline{1,m};\ j=\overline{1,n})\ \text{ yoki }\ A=\left\|a_{ij}\right\|\ \ (i=\overline{1,m};\ j=\overline{1,n}) \qquad (1.2)$$

ko`rinishda ham yoziladi.

Matritsada $m=n$ bo`lsa, *kvadrat matritsa* deyiladi:

$$A=\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}. \qquad (1.3)$$

Bosh diagonalidagi elementlari birlardan va qolgan elementlari nollardan iborat bo`lgan kvadrat matritsa *birlik matritsa* deyiladi va E deb belgilanadi. Masalan,

$$E=\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Matritsaning barcha elementlari nollardan iborat bo`lsa, *nol matri*

deb belgilanadi.

Mos elementlari teng, ya'ni $a_{ij} = b_{ij}$ bo`lgan bir xil o`lchamli A va B matritsalar *teng matritsalar* deyiladi.

Matritsaning satrlarini mos ustunlariga almashtirishdan hosil bo`lgan matritsa *transponirlangan matritsa* deyiladi va A^T kabi belgilanadi.

Masalan, $A = \begin{bmatrix} 3 & 2 \\ -4 & 5 \end{bmatrix}$ matritsa berilgan bo`lsa, A^T matritsani hisoblash uchun satrlarini mos ustunlariga almashtiramiz: $A^T = \begin{bmatrix} 3 & -4 \\ 2 & 5 \end{bmatrix}$.

Bir xil o`lchovli $A$ va $B$ matritsalarni qo`shish va ayirish mumkin.

Bir xil o`lchovli $A$ va $B$ *matritsalarni yig`indisi*(*ayirmasi*) deb shunday C matritsaga aytiladiki, uning elementlari A va B matritsalarning mos elementlari yig`indisiga(ayirmasiga) teng. $C = A + B$ $(C = A - B)$ kabi belgilanadi.

A matritsani λ *songa ko`paytmasi* deb, barcha elementlarin ko`paytirishdan hosil bo`lgan B matritsaga aytiladi, $B = \lambda A$ kabi belgilana

Matritsalarni qo`shish va songa ko`paytirish quyidagi xossalarga ega:

1. $A + B = B + A.$
2. $A + Q = A.$
3. $\lambda(A + B) = \lambda A + \lambda B.$
4. $(\lambda + \mu)A = \lambda A + \mu A.$

1-Misol.

Agar $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \\ -3 \end{pmatrix}$ bo'lsa $a_1, a_2 a_3$ ni toping.

Yechish. Tenglikning chap tomonidagi birinchi va ikkinchi matrisalarni qo'shib va yig'indini o'ng tomondagi matrisaga tenglashtirsak,

$$\begin{pmatrix} a_1 + 2 \\ a_2 + 3 \\ a_3 - 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \\ -3 \end{pmatrix}$$

Matrisalarning tengligi ta'rifidan foydalanib,

$a_1+2=7;\ a_2+3=8;\ a_3-1=-3,$ bu yerdan $a_1=5;\ a_2=5;\ a_3=-2.$

2-Misol. $A=\begin{pmatrix}-1 & 2 & 3 & -2\\ 3 & -3 & 2 & 0\end{pmatrix}$ matrisaga qarama-qarshi matrisani hamda $2A+3A=5A$ tenglikni tekshiring.

Yechish. Mustaqil bajaring.

3-misol. $A=\begin{bmatrix}2 & -1 & 3\\ 4 & 0 & 5\end{bmatrix}$ $B=\begin{bmatrix}-3 & 1\\ -5 & 4\\ 0 & 2\end{bmatrix}$ matritsalar berilgan. A^T+2B va $A-B^T$ matritsalarni hisoblang.

$$\blacktriangleright\ A^T+2B=\begin{bmatrix}2 & 4\\ -1 & 0\\ 3 & 5\end{bmatrix}+2\cdot\begin{bmatrix}-3 & 1\\ -5 & 4\\ 0 & 2\end{bmatrix}=\begin{bmatrix}2+2\cdot(-3) & 4+2\cdot 1\\ -1+2(-5) & 0+2\cdot 4\\ 3+2\cdot 0 & 5+2\cdot 2\end{bmatrix}=\begin{bmatrix}-4 & 6\\ -11 & 8\\ 3 & 9\end{bmatrix};$$

$$A-B^T=\begin{bmatrix}2 & -1 & 3\\ 4 & 0 & 5\end{bmatrix}-\begin{bmatrix}-3 & -5 & 0\\ 1 & 4 & 2\end{bmatrix}=\begin{bmatrix}2-(-3) & -1-(-5) & 3-\\ 4-1 & 0-4 & 5-\end{bmatrix}$$

$$=\begin{bmatrix}5 & 4 & 3\\ 3 & -4 & 3\end{bmatrix}.\blacktriangleleft$$

4. Misol. $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ matrisa berilgan. $2A+X=Q$ shartni qanoatlantiruvchi X matrisani toping, bu yerda Q –nol matrisa.

Yechish. $2A+X=Q$ tenglikning ikkala tomoniga –X matrisani qo'shamiz va chiziqli amallar bajaramiz.

$(2A+X)+(-\mathrm{X})=Q+(-\mathrm{X});\ \ 2A+(X+(-\mathrm{X}))=Q+(-\mathrm{X});\ 2A=-\mathrm{X},$ bu yerdan $\mathrm{X}=-2\mathrm{A},$ ya'ni $X=-2\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}=\begin{pmatrix}-2 & -4\\ -6 & -8\end{pmatrix}.$

$m\times k$ o`lchovli $A$ matritsani $k\times n$ o`lchovli B matritsaga ko`paytmasi deb, shunday $m\times n$ o`lchovli C matritsaga aytiladiki, uning elementlari

$$c_{ij}=a_{i1}b_{1j}+a_{i2}b_{2j}+\ldots+a_{ik}b_{kj} \qquad (1.4)$$

tenglik bilan aniqlanadi. $A_{m\times k}\cdot B_{k\times n}=C_{m\times n}$ kabi belgilanadi.

Demak, birinchi matritsaning ustunlari soni ikkinchi matritsaning satrlari soniga teng bo`lgan holdagini ularni ko`paytirish mumkin. Umuman olganda, $A\cdot B$ ko`paytma mavjud bo`ganda $B\cdot A$ ko`paytma mavjud bo`lavermaydi. $B\cdot A$ ko`paytma mavjud bo`gan holda ham, umuman olganda, $AB\neq BA$.

Agar $A\cdot B=B\cdot A$ bo`lsa, A va B matritsalar *kommutativlanadigan* matritsalar deyiladi.

5. Misol. $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ va $B=\begin{pmatrix}0 & 4\\ 6 & 6\end{pmatrix}$ matrisalar kommutativ ekanligini ko'rsating.

Yechish. Ikkala matrisa ham kvadratik matrisa hisoblanadi va o'lchovlari bir xil, bundan AB va BA ko'paytirish amali aniqlangan.

$$AB=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}0 & 4\\ 6 & 6\end{pmatrix}=\begin{pmatrix}1\cdot 0+2\cdot 6 & 1\cdot 4+2\cdot 6\\ 3\cdot 0+4\cdot 6 & 3\cdot 4+4\cdot 6\end{pmatrix}=\begin{pmatrix}12 & 16\\ 24 & 36\end{pmatrix},$$

$$BA=\begin{pmatrix}0 & 4\\ 6 & 6\end{pmatrix}\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}=\begin{pmatrix}0\cdot 1+4\cdot 3 & 0\cdot 2+4\cdot 4\\ 6\cdot 1+6\cdot 3 & 3\cdot 4+6\cdot 4\end{pmatrix}=\begin{pmatrix}12 & 16\\ 24 & 36\end{pmatrix},$$

ya'ni $AB=BA$, demak A va B matrisalar kommutativ.

$$A=\begin{pmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{pmatrix}\text{matritsaning}\quad B=\begin{pmatrix}b_{11} & b_{12} & b_{13}\\ b_{21} & b_{22} & b_{23}\\ b_{31} & b_{32} & b_{33}\end{pmatrix}$$

matritsaga **ko'paytmasi** deb elementlari quyidagicha aniqlanuvchi $S=A*B$ matritsaga aytiladi

$$S=A*B=\begin{pmatrix}a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31} & a_{11}b_{12}+a_{12}b_{22}+a_{13}b_{32} & a_{11}b_{13}+a_{12}b_{23}+a_{13}b_{33}\\ a_{21}b_{11}+a_{22}b_{21}+a_{23}b_{31} & a_{21}b_{12}+a_{22}b_{22}+a_{23}b_{32} & a_{21}b_{13}+a_{22}b_{23}+a_{23}b_{33}\\ a_{31}b_{11}+a_{32}b_{21}+a_{33}b_{31} & a_{31}b_{12}+a_{32}b_{22}+a_{33}b_{32} & a_{31}b_{13}+a_{32}b_{23}+a_{33}b_{33}\end{pmatrix}$$

Matritsalarni bu xilda ko'paytirish **satrlarni ustunga** deb yuritiladi. Matritsalarni ko'paytirish qoidasi birinchi ko'payuvchining ustunlari soni ikkinchi ko'payuvchining satrlari soniga teng bo'lgan har qanday to'g'ri burchakli matritsalar uchun o'rinlidir.

6-misol. $A=\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 1 \end{pmatrix}$ va $B=\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ matritsalarning ko'paytmasi topilsin.

Yechish. $A*B$ ko'paytma mavjud, chunki A matritsaning ustunlari 2 ga teng, B matritsaning satrlari soni ham 2 ga teng.

$$A*B=\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 1 \end{pmatrix}*\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}=\begin{pmatrix} 1\cdot 2+1\cdot 1 & 1\cdot(-1)+1\cdot 2 \\ 2\cdot 2+1\cdot 1 & 2\cdot(-1)+1\cdot 2 \\ -1\cdot 2+1\cdot 1 & (-1)\cdot(-1)+1\cdot 2 \end{pmatrix}=\begin{pmatrix} 3 & 1 \\ 5 & 0 \\ -1 & 3 \end{pmatrix}.$$

$B*A$ ko'paytma mavjud emas, chunki V matritsaning ustunlari soni 2 ga, A matritsaning satrlari soni esa 3 ga teng. Bu misol umumiy holda ko'paytirish o'rni almashtirish xossasiga ega emasligini ko'rsatadi, ya'ni u $A*B\neq B*A$.

7. Misol.

$$A=\begin{pmatrix} \sqrt{2} & 0 & 1 & -3 \end{pmatrix},\ B=\begin{pmatrix} \sqrt{2} \\ \pi \\ 5 \\ 1 \end{pmatrix},\ C=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, D=\begin{pmatrix} 1 & 2 \\ 5 & 3 \end{pmatrix}, F=\begin{pmatrix} 1 & 0 & -1 \\ 3 & 2 & 0 \end{pmatrix}$$

matrisalar berilgan bo'lsa ularning qaysilari ustida ko'paytirish amali bajariladi va bu ko'paytirish amalini bajaring.

Yechish. Matrisalarning o'lchovi $A_{14}, B_{41}, C_{22}, D_{22}, F_{23}$. Ta'rifga ko'ra quyidagi matrisalar ustida ko'paytirish amali bajaridladi: AB, BA, CD, DC, CF, DF. Ko'paytirish amallarini mustaqil bajaring.

Matritsalarni ko`paytirish quyidagi xossalarga ega:

1. $(\lambda A)B=\lambda(AB)$.
2. $(A+B)C=AC+AB$.

3. $A(B+C)=AB+AC$.

4. $A(BC)=(AB)C$.

Keltirilgan xossalardan to‘rtinchisini quyidagi misol yordamida tekshiramiz.

Transponirlangan matritsa uchun esa quyidagi formula o`rinli:

1. $(A^T)^T=A$.

2. $(AB)^T=B^T\cdot A^T$.

Agar A kvadrat matritsaning determinanti noldan farqli bo`lsa, ya’ni $\det A\neq 0$ bo`lsa, A matritsa *xosmas matritsa* deyiladi.

8.Misol.

$A=\begin{pmatrix}2&3&-1\\0&5&4\\1&1&2\end{pmatrix}$, $B=\begin{pmatrix}-3&0\\1&2\\5&3\end{pmatrix}$, $C=\begin{pmatrix}1&-2\\2&3\end{pmatrix}$ matrisalarning ko’paytirishga nisbatan assosiativligidan foydalanib ABC ko’paytmani hisoblang.

Yechish. Matrisalar tartibi A_{33}, B_{32}, C_{22}. Berilgan matrisalarning turin holda, quyidagi ko’paytmalar aniqlangan $(AB)_{32}$, $(AB)C$, $(BC)_{32}$, $A(BC)$. bilan yechish mumkin.

1. $(AB)C$: $AB=\begin{pmatrix}2&3&-1\\0&5&4\\1&1&2\end{pmatrix}\begin{pmatrix}-3&0\\1&2\\5&3\end{pmatrix}=\begin{pmatrix}-8&3\\25&22\\8&8\end{pmatrix}$;

$$(AB)C=\begin{pmatrix}-8&3\\25&22\\8&8\end{pmatrix}\begin{pmatrix}1&-2\\2&3\end{pmatrix}=\begin{pmatrix}-2&25\\49&16\\24&8\end{pmatrix}.$$

2. $A(BC)$: $BC=\begin{pmatrix}-3&0\\1&2\\5&3\end{pmatrix}\begin{pmatrix}1&-2\\2&3\end{pmatrix}=\begin{pmatrix}-3&6\\5&4\\11&-1\end{pmatrix}$;

$$A(BC)=\begin{pmatrix}2&3&-1\\0&5&4\\1&1&2\end{pmatrix}\begin{pmatrix}-3&6\\5&4\\11&-1\end{pmatrix}=\begin{pmatrix}-2&25\\49&16\\24&8\end{pmatrix}.$$

Tabiiyki, ikkala usulda ham bir xil natija beradi.

9. Мисол. $\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}^9$ matritsani toping.

Yechish. $A=\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$ belgilash kiritib A matritsaning kvadratini hisoblaymiz: $A^2=\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}\cdot\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. $A^2=E$ - birlik matritsa. Shuning uchun $A^9=\left(A^2\right)^4\cdot A=E^4A=A$ va $\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}^9=\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$.

10-misol.

$A=\begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$, $B=\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$ matritsalar berilgan. $(AB)^T=B^T\cdot A^T$. Tenglikni tekshiring

$$A\cdot B=\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}\cdot\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}=\begin{bmatrix} 2\cdot 3+2\cdot 2 & 2\cdot 4+2\cdot 3 \\ 1\cdot 3+2\cdot 2 & 1\cdot 4+2\cdot 3 \end{bmatrix}=\begin{bmatrix} 10 & 14 \\ 7 & 10 \end{bmatrix};$$

$(A\cdot B)^T=\begin{pmatrix} 10 & 7 \\ 14 & 10 \end{pmatrix}$; Endi A^T va B^T larni aniqlaymiz.

$$A^T=\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix},\ B^T=\begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix};\ B^TA^T=\begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}=\begin{pmatrix} 10 & 7 \\ 14 & 10 \end{pmatrix}.$$

11. Мисол. Agar $f(x)=-2x^2+5x+9$, $A=\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$ bo‘lsa, $f(A)$ matritsali ko‘phadning qiymatini toping.

Yechish. $A^2=A\cdot A=\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}\cdot\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}=\begin{pmatrix} 1\cdot 1+2\cdot 3 & 1\cdot 2+2\cdot 0 \\ 3\cdot 1+0\cdot 3 & 3\cdot 2+0\cdot 0 \end{pmatrix}=\begin{pmatrix} 7 & 2 \\ 3 & 6 \end{pmatrix}$,

$$f(A)=-2A^2+5A+9E=-2\cdot\begin{pmatrix} 7 & 2 \\ 3 & 6 \end{pmatrix}+5\cdot\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}+9\cdot\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}=$$

$$=\begin{pmatrix} -14 & -4 \\ -6 & -12 \end{pmatrix}+\begin{pmatrix} 5 & 10 \\ 15 & 0 \end{pmatrix}+\begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix}=\begin{pmatrix} 0 & 6 \\ 9 & -3 \end{pmatrix}.$$

Auditoriya topshiriqlari

1. Quyidagi matrisalar tengligidan $x, y, z \in R$ ni hisoblang.

$$\begin{pmatrix} x & 2x+y \\ x+z & 2y+z \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 4 & -22 \end{pmatrix}.$$

J.: $x = 6,\ y = -10,\ z = -2.$

2. Quyidagi mulohazalardan qaysi biri rost?

$$a)\begin{pmatrix} 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix};\ b)\ \begin{pmatrix} 0-0 & 0 \end{pmatrix} = \begin{pmatrix} 0-0 & 0-0 \end{pmatrix};\ \text{c)}\ \begin{pmatrix} 1-1 & 2-2 \\ x-xz & 3-3 \end{pmatrix} = \begin{pmatrix} 0 \end{pmatrix},\ x \in R$$

J.: $b) - \text{rost};\ a)\ va\ c) - \text{yolg'on}.$

3. $A,\ B$ va C matrisalar berilgan.

$$A = \begin{pmatrix} 5 & 2 & 1 \\ 3 & 0 & 2 \\ -1 & 8 & 4 \end{pmatrix};\ B = \begin{pmatrix} 0 & 4 & 1 \\ 2 & 6 & 3 \\ 1 & -2 & 1 \end{pmatrix};\ C = \begin{pmatrix} 2 & 0 & 6 \\ 1 & 1 & 3 \\ 2 & 5 & 6 \end{pmatrix}.$$

Quyidagilarni hisoblang.

$a)\ 2A - 3B - C;\ b)\ A + 2B - 3C;\ c)\ A^T + 2B + C^T;\ d)\ 2A - B^T + 2C^T.$

4. Quyidagi matrisalar uchun umumiy ko'paytmani qavsdan tashqariga ch

$$a)\ A = \begin{pmatrix} 0.00920 & 0.00011 & 0 \\ 0.00034 & -0.00217 & 0.00024 \end{pmatrix};\ \text{b)}\ B = \begin{pmatrix} x^2 & x^3 y & 2x \\ x^3 - x & 3x^2 & x^2 - x \end{pmatrix};\ \text{c)}\ C = \begin{pmatrix} \frac{1}{5} & \frac{3}{20} \\ \frac{-2}{15} & \frac{5}{12} \end{pmatrix}.$$

J.: $a)\ A = 10^{-5}\begin{pmatrix} 920 & 11 & 0 \\ 34 & -217 & 24 \end{pmatrix};\ \text{b)}\ \text{B} = \text{x}\begin{pmatrix} x & x^2 y & 2 \\ x^2 - 1 & 3x & x - 1 \end{pmatrix};\ \text{c)}\ \text{C} = \frac{1}{60}\begin{pmatrix} 12 & 9 \\ -8 & 25 \end{pmatrix}.$

5. Agar $X + Y = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$ va $X - Y = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$ bo'lsa, X va Y matrisalarni aniqlang.

J.: $X = \frac{3}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},\quad Y = \frac{1}{2}\begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$

6. $A=\begin{pmatrix}3 & 6\\ 6 & 9\end{pmatrix}$ matrisa berilgan. Quyidagi o'artlarni qanoatlantiruvchi X matrisani toping: $a)\ 2A-X=0;\ b)\ 3A+X=0;\ c) 2A+3X=0.$

7. $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ matrisa berilgan. Quyidagi o'artlarni qanoatlantiruvchi X matrisani toping: $a)\ A+X=E;\ b)\ 3A-2X=E,$ bu yerda $E-$ birlik matrisa.

8. Agar $A_{23}\cdot B_{34}=C_{ml}$ bo'lsa, S matrisaning o'lchovi $m\ va\ l$ ni toping.

J.: $m=2,\ l=4.$

9. Agar $A_{34}\cdot B_{nl}=C_{35}$ ekanligi ma'lum bo'lsa, $n\ va\ l$ ni toping.

J.: $n=4,\ l=5.$

10. Agar $A_{4n}\cdot B_{k5}=C_{45}$ bo'lsa, S matrisaning o'lchovi $n\ va\ k$lar o'zaro qanday bog'langan.

11. Quyidagi matrisalar berilgan

$$A=\begin{pmatrix}1 & -3\\ 2 & -4\end{pmatrix};\ B=\begin{pmatrix}3 & 2 & -1\\ 5 & 4 & -7\end{pmatrix};\ C=\begin{pmatrix}1 & 0 & 2\\ 3 & 1 & 5\\ 6 & 4 & 8\end{pmatrix};\ \mathrm{D}=\begin{pmatrix}4\\ 9\\ 6\end{pmatrix}.$$

Quyidagi ko'paytmalarning qaysilari mavjud.

1)AB; 2)BA; 3)BC; 4)CB; 5)CD; 6)DC; 7)AC; 8)CA; 9)BD; 10)DB?

J.:1); 3); 5); 9) -mavjud; 2); 4); 6); 7); 8); 10) mavjud emas.

12. AB va BA matrisalar ko'paytmasini toping

$$A=\begin{pmatrix}1 & 2 & -3\\ 4 & 0 & -1\end{pmatrix};\ B=\begin{pmatrix}2 & 1\\ 3 & 4\\ 0 & 5\end{pmatrix}.$$

J.: $AB=\begin{pmatrix}8 & -6\\ 8 & -1\end{pmatrix};\ BA=\begin{pmatrix}6 & 4 & -7\\ 19 & 6 & -13\\ 20 & 0 & -5\end{pmatrix}.$

13. $A=\begin{pmatrix}1 & 0 & -4 & 8\\ 2 & 5 & -3 & 0\\ 3 & 0 & -2 & 9\\ 4 & 6 & -1 & 0\end{pmatrix}$; $B=\begin{pmatrix}0 & 8\\ 9 & 6\\ 5 & 4\\ 1 & 2\end{pmatrix}$ matrisalar berilgan. $C=AB$ matrisaning uchinchi satr ikkinchi ustundagi elementni toping.

J.: $c_{32}=34.$

14. A va B matrisalar uchun $(A+3B)^2$ ni toping, agar

$$A=\begin{pmatrix}1 & 4 & 7\\ 2 & 5 & -8\\ -3 & 6 & 9\end{pmatrix};\quad B=\begin{pmatrix}-2 & 1 & -1\\ 1 & 0 & 2\\ 4 & -1 & 0\end{pmatrix}.$$

J.: $\begin{pmatrix}96 & 12 & 8\\ -18 & 54 & -8\\ 51 & 85 & 111\end{pmatrix}.$

15. $A=\begin{pmatrix}5 & 9 & 7\\ 0 & 3 & -2\end{pmatrix}$; $B=\begin{pmatrix}5 & 9\\ 0 & 3\\ 0 & 2\end{pmatrix}$; $C=\begin{pmatrix}3 & 4\\ -1 & 0\end{pmatrix}$. $(AB)C$ va $A(BC)$ ni aniqla

J.: $\begin{pmatrix}-5 & 0\end{pmatrix}.$

16. Matrisalar ko'paytmasini toping.

$$\begin{pmatrix}1 & -2 & 1\end{pmatrix}\begin{pmatrix}2 & -3 & 1\\ 1 & 2 & -3\\ 0 & 1 & 4\end{pmatrix}\begin{pmatrix}1\\ 0\\ 2\end{pmatrix}.$$

J.: (22).

17. $A=A_{mn}$ va $B=B_{lk}$ matrisalar berilgan. Qaysi holatda ko'paytma aniqlangan.

1) AB; 2) BA; 3) AA^T; 4) AB^T; 5) BA^T; 6) $A^T A$; 7) $A^T B^T$; 8) $B^T A^T$;

J.: 1) $n=l$; 2) $m=k$; 3) doimo; 4) $n=k$; 5) $n=k$; 6) doimo; 7) $m=k$; 8) $n=l$

18. A va B matritsalar berilgan. $A+B$, $2A-B$ va $A+3B$ matritsalarni toping.

a) $A=\begin{bmatrix}3 & 0 & -1\\ 5 & 4 & -2\end{bmatrix}$, $B=\begin{bmatrix}-1 & 3 & 4\\ 0 & 2 & 5\end{bmatrix}$;

b) $A=\begin{bmatrix}2 & -4\\ 1 & -1\\ 5 & -3\end{bmatrix},\ B=\begin{bmatrix}-3 & 2\\ -2 & 4\\ 0 & 5\end{bmatrix}.$

19. A va B matritsalar berilgan. AB va BA matritsalarni toping.

a) $A=\begin{bmatrix}1 & -2 & 4\\ -3 & 5 & 0\\ 2 & -1 & 3\end{bmatrix},\ B=\begin{bmatrix}3 & 2 & 5\\ 4 & 0 & -1\\ 1 & 1 & -3\end{bmatrix};$

b) $A=\begin{bmatrix}3 & 2 & -1\\ 7 & 4 & -2\end{bmatrix},\ B=\begin{bmatrix}0 & -4\\ 1 & -1\\ 5 & -3\end{bmatrix};$

d) $A=\begin{bmatrix}2 & -5 & 3 & 0\end{bmatrix},\ B=\begin{bmatrix}4\\ 2\\ -3\\ 5\end{bmatrix};$

e) $A=\begin{bmatrix}11 & 10\\ 2 & 9\end{bmatrix},\ B=\begin{bmatrix}3 & 5\\ 1 & 2\end{bmatrix}.$

20. A, B va C matritsalar berilgan. $(AB)C=A(BC)$ ekanini tekshiring.

$$A=\begin{bmatrix}5 & 3\\ 2 & -1\\ 4 & 7\end{bmatrix},\quad B=\begin{bmatrix}4 & -3\\ -5 & 2\end{bmatrix},\quad C=\begin{bmatrix}3 & 5\\ 1 & 2\end{bmatrix}.$$

21. Matritsalarni hisoblang:

1. $\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}^5.$ **2.** $\begin{pmatrix}3 & 2\\ -4 & -2\end{pmatrix}^5.$ **3.** $\begin{pmatrix}1 & -2\\ 3 & -4\end{pmatrix}^3.$

22. $f(A)$ matritsali ko‘phadning qiymatini toping:

1). $f(x)=3x^3+x^2+2,\ A=\begin{pmatrix}1 & 5\\ 0 & -3\end{pmatrix}.$

2). $f(x)=2x^3-3x^2+5,\ A=\begin{pmatrix}1 & 2\\ -2 & 3\end{pmatrix}.$

3). $f(x)=3x^2-5x+2,\ A=\begin{pmatrix}1 & 2 & 0\\ 0 & 2 & -1\\ -2 & 1 & 4\end{pmatrix}.$

4). $f(x)=x^3-6x^2+9x+4,\ A=\begin{pmatrix}1 & 0 & 0\\ 0 & 2 & -1\\ 0 & 1 & 4\end{pmatrix}.$

5). $f(x)=x^2+x+1,\ A=\begin{pmatrix}2 & 1 & 1\\ 1 & 2 & 1\\ 1 & 1 & 2\end{pmatrix}.$

Mustaqil yechish uchun testlar

1. $A=\begin{pmatrix}5 & 2\\ 1 & 4\end{pmatrix}$ va $B=\begin{pmatrix}-3 & 1\\ 5 & -2\end{pmatrix}$ matritsa berilgan, $A+2B$ matritsani toping.

A) $\begin{pmatrix}-1 & 4\\ 11 & 0\end{pmatrix}$, B) $\begin{pmatrix}-5 & 1\\ 17 & -7\end{pmatrix}$, D) $\begin{pmatrix}-1 & 4\\ 8 & 0\end{pmatrix}$, E) $\begin{pmatrix}-11 & 4\\ 11 & 0\end{pmatrix}$.

2. $K=\begin{pmatrix}1 & -2 & 3\\ -2 & 3 & -4\end{pmatrix}$ matritsa berilgan bo‘lsa, $2K$ matritsani toping.

A) $\begin{pmatrix}2 & -4 & 6\\ -2 & 3 & -4\end{pmatrix}$, B) $\begin{pmatrix}1 & -2 & 3\\ -4 & 6 & -8\end{pmatrix}$, D) $\begin{pmatrix}2 & -4 & 6\\ -4 & 6 & -8\end{pmatrix}$, E) A va B to‘g‘ri

3. $A=\begin{pmatrix}5 & 2\\ 1 & 4\end{pmatrix}$ va $B=\begin{pmatrix}-3 & 1\\ 5 & -2\end{pmatrix}$ matritsa berilgan, matritsani toping.

A) $\begin{pmatrix}-13 & 7\\ 25 & -7\end{pmatrix}$, B) $\begin{pmatrix}-5 & 7\\ 8 & 10\end{pmatrix}$, D) $\begin{pmatrix}-5 & 1\\ 17 & -7\end{pmatrix}$, E) $\begin{pmatrix}-10 & 2\\ 5 & -2\end{pmatrix}$.

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